# ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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**C01-4**

**EXPERIMENT 4**

**Aim:**

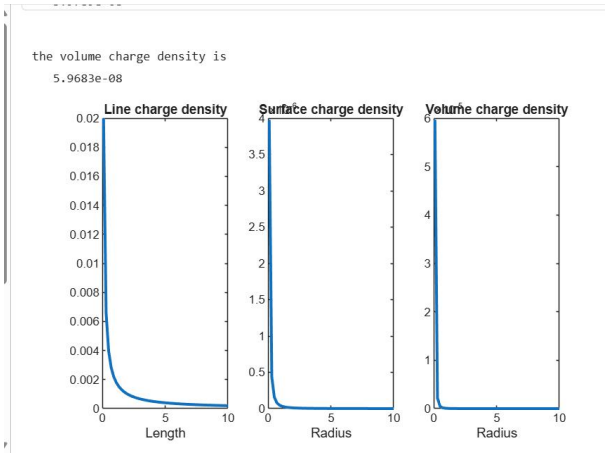
To write program for continuous Charge Distribution in electric field. The different types of charge density are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using MATLAB.

**Test case (i) Determination of line, surface and volume charge density for given charge, length and radius**

**PROGRAM:**

```
clc;
clear;
q=input("Enter the charge value is:");
l=input("Enter the length value is:");
% Line charge density
rho_l = q/l;
disp('the line charge density is');
disp(rho_l);
% Surface charge density
r=input("Enter the radius value is:");
rho_s = q/(4*pi*r^2);
disp('the surface charge density is');
disp(rho_s);
% Volume charge density
rho_v = q/((4/3)*pi*r^3);
disp('the volume charge density is');
disp(rho_v);
% Plot
x = linspace(0.1,10,50);
y = rho_l./x;
z = rho_s./(x.^2);
v = rho_v./(x.^3);
figure;
subplot(1,3,1);
plot(x,y,'LineWidth',2);
title('Line charge density');
xlabel('Length');
subplot(1,3,2);
plot(x,z,'LineWidth',2);
title('Surface charge density');
xlabel('Radius');
subplot(1,3,3);
plot(x,v,'LineWidth',2);
title('Volume charge density');
xlabel('Radius');
```

**OUTPUT:**

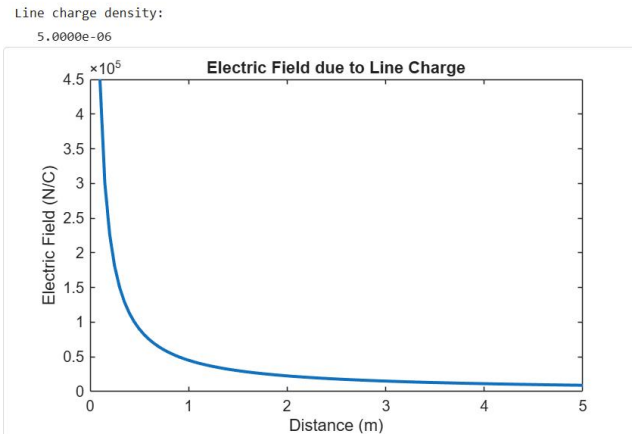



**Test case (ii) A long wire of length L = 10 m carries a uniform charge of Q = 50 µC. Calculate the line charge density and plot electric field variation**

**Program:**

```
clc;
L = 10;
Q = 50e-6;
lambda = Q/L;
disp("Line charge density:");
disp(lambda);
r = linspace(0.1,5,100);
k = 9e9;
E = k*lambda./r;
plot(r,E,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field due to Line Charge');
```

**OUPUT:**



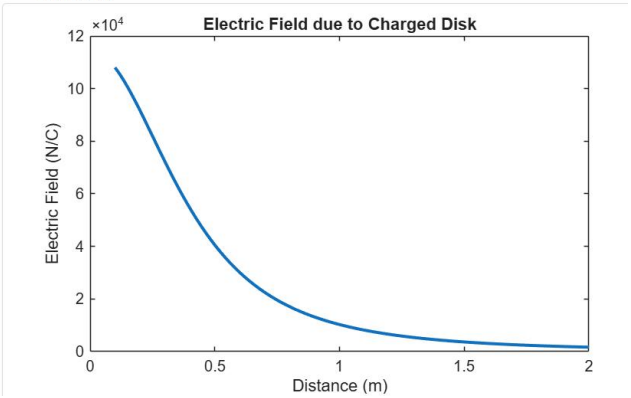
**Test case (iii) A uniformly charged disk of radius R = 0.5 m with total charge Q = 10 µC. Calculate surface charge density and electric field**

**Program:**

```
clc;
R = 0.5;
Q = 10e-6;
A = pi*R^2;
sigma = Q/A;
disp("Surface charge density:");
disp(sigma);
z = linspace(0.1,2,100);
k = 9e9;
E = (k*sigma*R^2)./(2*(z.^2 + R^2).^(3/2));
plot(z,E,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field due to Charged Disk');
```

**OUTPUT:**

```
Surface charge density:
   1.2732e-05
```



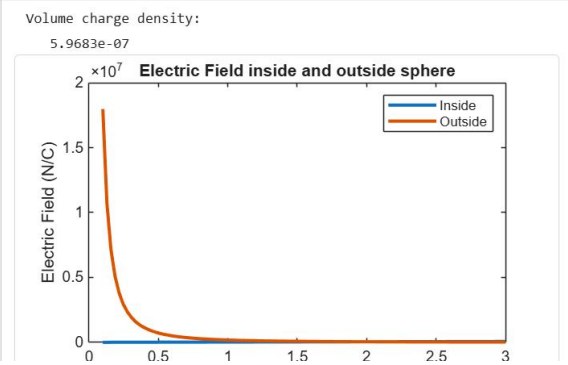

**Test case (iv) A uniformly charged sphere of radius R = 2 m with total charge Q = 20 µC. Calculate volume charge density and electric field inside and outside 3**

**Program:**

```
clc;
R = 2;
Q = 20e-6;
V = (4/3)*pi*R^3;
rho = Q/V;
disp("Volume charge density:");
disp(rho);
r = linspace(0.1,3,100);
epsilon0 = 8.854e-12;
k = 1/(4*pi*epsilon0);
E_inside = (rho.*r)/(3*epsilon0);
E_outside = k*Q./(r.^2);
plot(r,E_inside,'LineWidth',2); hold on;
plot(r,E_outside,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field inside and outside sphere');
```

legend('Inside','Outside');

**OUTPUT:**




**RESULTS:**

The different types of charge density are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density has been done using MATLAB.

| **Objective** | **Assessment Criterion** | **Excellent – 4** | **Good – 3** | **Satisfactory – 2** | **Needs Improvement – 1** | **Weightage** | **Marks given** | **Marks Obtained** |
|---|---|---|---|---|---|---|---|---|
| Identify physical quantities and force models | Identify physical quantities and force models | Correctly identifies charge/current, separation, medium and force direction for both cases. | Identifies most quantities and models correctly. | Needs guidance in selecting the correct relation or direction. | Uses incorrect force models or quantities. | 25 | 2.5 | |
| Calculate electric and magnetic forces | Calculate electric and magnetic forces | Implements Coulomb-force and parallel-current force calculations correctly with units. | Calculations are mostly correct with minor errors. | Some calculations are correct but require assistance. | Force calculations are invalid or absent. | 25 | 2.5 | |
| Evaluate force versus separation | Evaluate force versus separation | Correctly varies separation, plots trends and checks | Trend and plots are correct with | Limited sweep or incomplete trend interpretation. | Does not establish a valid relationship. | 30 | 3 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | theoretical distance dependence. | minor omission2s. | | | | | |
| Explain and report results | Explain and report results | Explains attraction/repulsion and reports assumptions, units and conclusions clearly. | Conclusion is correct with minor documentation gaps. | Conclusion is incomplete or partly inaccurate. | Conclusion is missing or incorrect. | 20 | 2 | |
| **Total** | | | | | | 100 | 10 | |